

SUR, Oman

WMO: 412680

Lat: 22.533N

Lon: 59.483E

Elev: 14

StdP: 101.16

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.3	17.3	0.8	4.0	29.8	3.1	4.7	30.2	11.5	23.9	10.5	24.4	2.3	230	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	9.2	43.4	24.5	42.1	24.3	40.9	24.3	30.4	36.8	29.5	35.8	28.7	35.0	6.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
28.9	25.6	34.9	27.9	24.1	34.1	27.0	22.8	33.6	102.5	36.8	97.9	36.2	93.6	35.0	36.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.1	12.7	11.2	DB	13.8	46.2	0.7	1.1	13.4	47.0	13.0	47.6	12.6	48.2	12.1	49.0	
			WB	9.4	30.5	1.2	2.0	8.5	31.9	7.8	33.1	7.1	34.3	6.2	35.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	29.6	22.0	23.5	26.6	31.2	35.5	36.0	34.3	33.2	32.6	30.6	26.1	23.1	(6)
		DBStd	5.36	1.72	2.43	2.88	2.94	2.53	2.38	2.47	2.49	1.90	2.10	1.86	1.63	
		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
		CDD10.0	7146	372	377	516	636	789	779	754	718	677	637	483	406	
		CDD18.3	4104	114	144	257	386	531	529	496	460	427	379	233	147	
		CDH23.3	54558	403	844	2458	5313	8554	8632	7726	6819	6032	4937	2092	747	
(13)		CDH26.7	33288	42	209	983	3080	6077	6233	5250	4363	3664	2648	654	84	(13)
(14)	Wind	WSAvg	4.9	3.4	4.2	4.4	4.7	6.1	6.3	6.4	6.1	5.8	4.2	3.2	3.3	(14)
(15)	Precipitation	PrecAvg	95	12	24	15	9	0	4	3	3	0	2	8	9	(15)
(16)		PrecMax	255	99	197	104	41	4	30	28	38	0	27	67	84	(16)
(17)		PrecMin	3	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	76	22	49	26	13	1	9	8	8	0	6	18	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.4	33.1	37.5	41.5	44.8	45.2	44.4	43.5	41.3	39.2	34.5	30.4	(19)
(20)			MCWB	19.6	18.7	19.5	22.3	25.0	23.9	24.9	25.7	23.6	20.9	20.5	20.3	(20)
(21)		2%	DB	27.7	30.7	35.1	39.4	42.9	43.5	42.2	40.9	39.4	37.3	32.6	28.6	(21)
(22)			MCWB	19.5	19.1	19.1	21.7	23.6	24.6	25.0	25.2	24.2	21.9	20.6	20.3	(22)
(23)		5%	DB	26.6	29.1	33.2	37.7	41.5	42.1	40.6	39.1	38.0	36.0	31.2	27.4	(23)
(24)			MCWB	19.4	19.0	19.2	21.3	23.2	24.7	25.2	25.3	24.3	22.1	21.3	20.2	(24)
(25)		10%	DB	25.4	27.6	31.4	36.1	40.2	40.8	38.9	37.6	36.5	34.7	29.9	26.5	(25)
(26)			MCWB	19.2	19.0	19.2	20.9	23.1	25.1	25.6	25.5	24.3	22.1	21.8	20.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.2	23.5	24.9	28.0	31.6	31.6	31.0	30.9	30.5	29.1	26.2	23.9	(27)
(28)			MCDB	25.9	27.2	28.8	37.1	40.2	38.8	37.3	35.6	35.5	33.1	29.9	26.8	(28)
(29)		2%	WB	22.0	22.4	24.0	26.4	29.5	30.2	29.7	29.5	29.0	27.4	24.9	22.8	(29)
(30)			MCDB	25.2	26.5	28.1	33.4	36.7	37.9	35.5	35.0	33.9	31.7	28.5	26.2	(30)
(31)		5%	WB	21.1	21.5	23.1	25.2	28.2	29.3	28.7	28.3	27.9	26.3	24.1	21.9	(31)
(32)			MCDB	24.5	25.8	27.8	32.1	35.2	36.3	34.7	34.8	33.4	31.3	28.0	25.6	(32)
(33)		10%	WB	20.1	20.7	22.3	24.2	26.9	28.3	27.8	27.4	26.9	25.3	23.3	21.1	(33)
(34)			MCDB	24.1	25.2	27.3	30.8	34.1	35.1	34.4	34.3	32.7	30.8	27.6	25.3	(34)
(35)	Mean Daily Temperature Range	MDBR	7.1	7.9	8.1	8.3	9.0	9.2	8.7	8.9	9.2	8.7	7.7	7.1		(35)
(36)		5% DB	MCDBR	8.4	9.5	10.4	10.1	10.4	11.3	11.4	10.9	10.8	10.0	9.2	8.1	(36)
(37)			MCWBR	3.6	3.8	4.8	4.3	4.2	4.1	3.6	3.8	3.8	4.8	4.3	3.3	(37)
(38)		5% WB	MCDBR	7.2	7.7	7.7	8.5	8.9	8.7	8.1	8.1	8.5	7.9	7.4	7.1	(38)
(39)			MCWBR	3.9	4.1	4.7	4.6	4.8	4.6	5.0	5.1	5.1	5.3	4.3	3.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.428	0.471	0.536	0.602	0.653	0.740	0.815	0.750	0.621	0.487	0.451	0.427		(40)
(41)		taud	2.175	2.029	1.854	1.741	1.654	1.522	1.447	1.532	1.717	1.988	2.111	2.173		(41)
(42)		Ebn at Noon	828	814	779	735	694	631	586	625	701	788	795	811		(42)
(43)		Edh at Noon	135	166	207	235	255	289	311	286	234	171	143	131		(43)
(44)	All-Sky Solar Radiation	RadAvg	4.44	5.24	6.03	6.61	7.05	6.62	6.11	6.14	6.00	5.53	4.64	4.22		(44)
(45)		RadStd	0.18	0.20	0.28	0.20	0.11	0.28	0.31	0.15	0.10	0.14	0.17	0.18		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+0.23	N/A	N/A	N/A	+0.39	+0.45	N/A	N/A	+96	+98

Nomenclature: See separate page